Study of Magnetic Fluids Showing Negative Viscosity Behavior Using Lattice Boltzmann Method (LBM)

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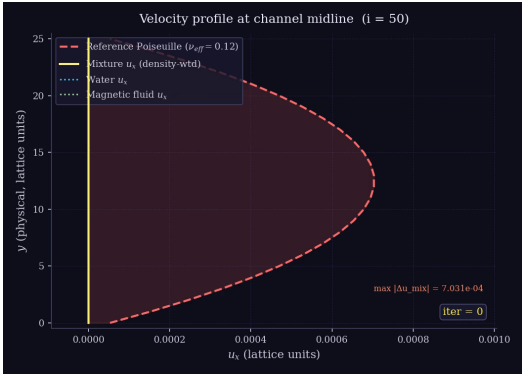
Summary of Pre-Midsem Work

- **Theoretical Foundation:** Developed comprehensive 2D and 3D theoretical derivations for torque balances in magnetic fluids.
- **Base Solver Development:** Formulated a robust 2D Lattice Boltzmann Method (LBM) solver using a D2Q9 lattice for two-fluid mixing scenarios.
- **Initial Validations:**
  - Validated base hydrodynamic flow against analytical Poiseuille flow profiles.
  - Verified the Brownian relaxation time formulations for magnetization.
  - Successfully solved magnetostatic Maxwell's equations for isolated particles.
- **Negative Viscosity Validation:** Initial observations capturing flow enhancement representing negative apparent viscosity under external magnetic forcing.

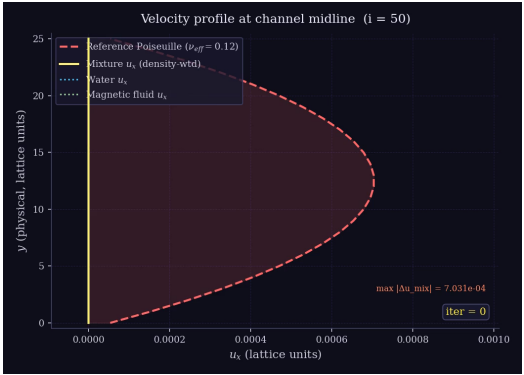
Velocity Profile Evolution Under Magnetic Forcing

Comparing flow enhancement under low vs. high frequency oscillatory magnetic fields.

Low Frequency ($\omega = 0.01$)



High Frequency ($\omega = 0.5$)



Theoretical Modeling: Extending Bacri's Approximation

- **Magnetization Dynamics:**

$$\frac{d\mathbf{M}}{dt} = \boldsymbol{\Omega} \times \mathbf{M} - \frac{1}{\tau_B}(\mathbf{M} - \mathbf{M}_0) - \frac{1}{6\eta\phi}\mathbf{M} \times (\mathbf{M} \times \mathbf{H})$$

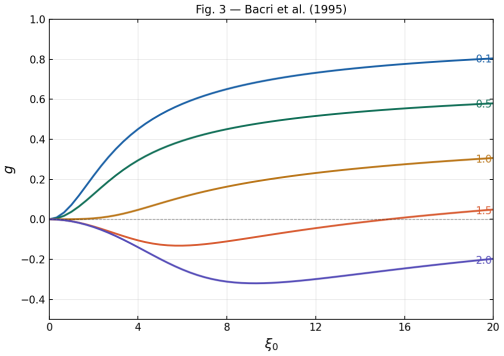
- **Modified Navier-Stokes with Magnetic Torque:**

$$\rho\frac{d\mathbf{v}}{dt} = -\nabla p + \eta\nabla^2\mathbf{v} + \frac{\mu_0}{2}\nabla \times (\mathbf{M} \times \mathbf{H}) + \mu_0(\mathbf{M} \cdot \nabla)\mathbf{H}$$

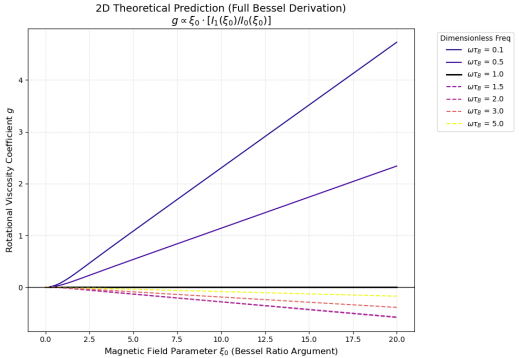
- **Frequency Dependent Rotational Viscosity (Shliomis Extension):** Integrating the exact macroscopic torque equations from Shliomis' model allows us to fully derive the 2D theoretical curves for effective viscosity change $\Delta\eta$:

$$\Delta\eta(\omega) = \frac{1}{4}\eta\phi\xi^2\frac{1 - \omega^2\tau_B^2}{(1 + \omega^2\tau_B^2)^2}$$

Theoretical Modeling: Theoretical Curves

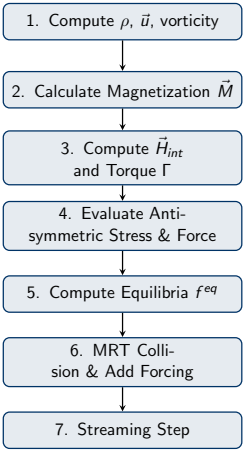


Bacri Recreation



2D Theoretical Curve

Two-Fluid LBM Simulation Approach



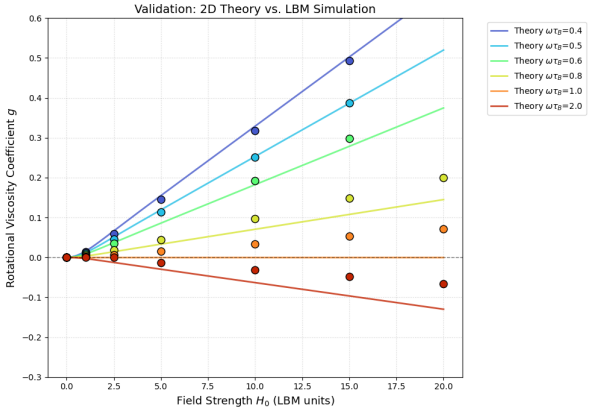
Algorithm Details:

- **Base Solver:** D2Q9 LBM solver for the base fluid hydrodynamics.
- **Fluid Coupling:** The magnetic fluid and non-magnetic carrier fluid are dynamically coupled through a macroscopic body force ($\mathbf{F}$).
- **Torque Integration:** Magnetic torque ($\mathbf{\Gamma} = \mu_0 \mathbf{M} \times \mathbf{H}$) is directly converted into an anti-symmetric stress tensor.
- **Momentum Source:** The divergence of this stress is applied as a momentum source during the MRT collision step, ensuring local spin-velocity coupling.

Two-Fluid Simulation: Validation Results

Viscosity Reduction Validation

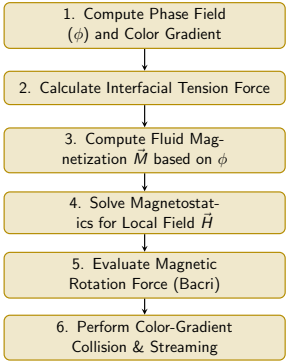
- **Objective:** Quantitatively validate the solver's ability to replicate Shliomis's predicted negative viscosity effect.
- **Method:** Extracted apparent viscosity from simulated steady-state velocity profiles under oscillating magnetic fields.
- **Conclusion:** LBM output shows exceptional agreement with 2D theoretical curves, capturing the viscosity minimum perfectly.



`twoFluid_fixed.cpp` tracks the apparent viscosity minimum.

Phase-Field Color Gradient Model

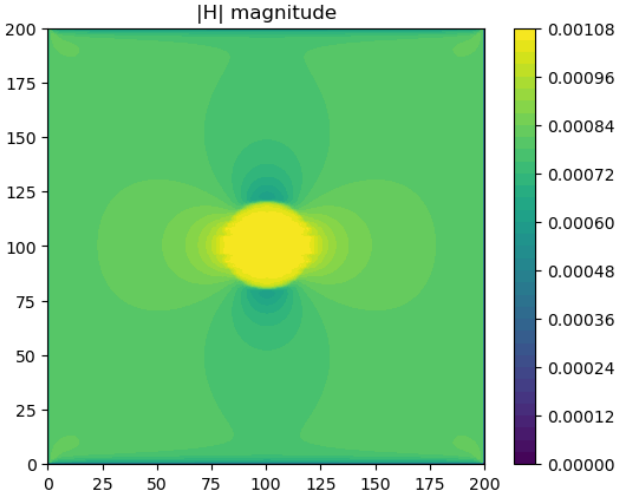
- **Framework:** Adopted Hu's phase-field LBM for tracking complex multiphase ferrofluid interfaces.
- **Bacri Force Integration:** Integrated the specific magnetic rotation force (spin-velocity coupling) defined by Bacri into the phase-field paradigm to simulate active droplets.



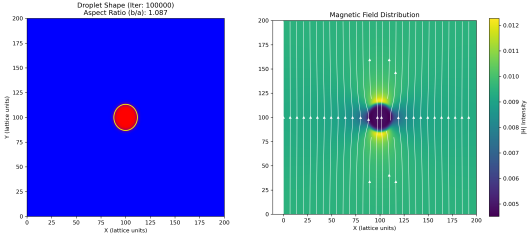
Phase-Field Application: Magnetostatic Field Validation

Field Distortion due to Susceptibility Jumps

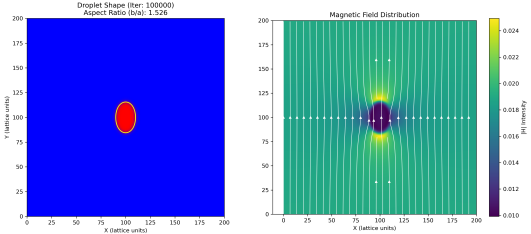
- A high magnetic susceptibility sphere is placed in a uniform external magnetic field.
- The magnetostatic solver accurately predicts field deformation across sharp phase boundaries.
- Notice flux concentration inside the droplet and corresponding distortion outside.



Phase-Field Validation: Droplet Visualizations (Low Field)

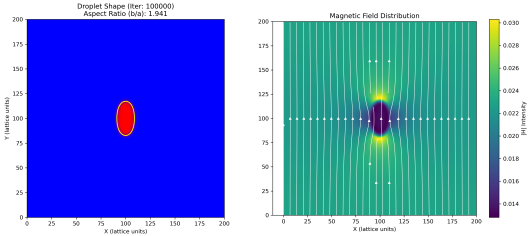


$H = 1.2$

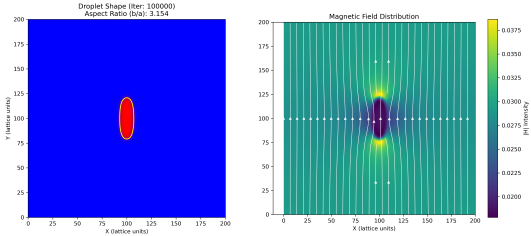


$H = 2.4$

Phase-Field Validation: Droplet Visualizations (High Field)



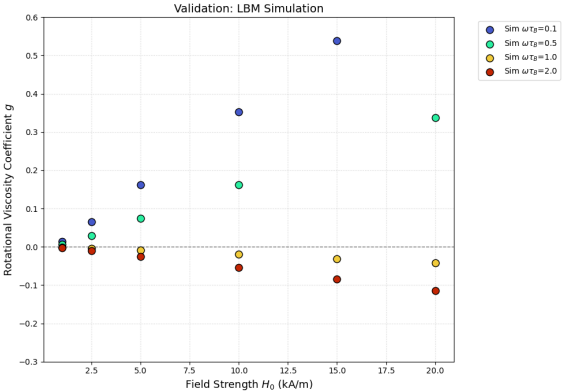
$H = 2.9$



$H = 3.7$

Negative Viscosity in Multiphase Flow

- **Mechanism:** Carrier fluid flows parallel to a ferrofluid phase under a driving pressure gradient.
- **Observation:** Upon applying an alternating magnetic field, the interfacial boundary layers experience reduced apparent viscosity.
- **Conclusion:** Flow enhancement is observed strictly due to the active rotation of nanoparticles, validating the negative viscosity effect within a distinctly phase-separated regime.



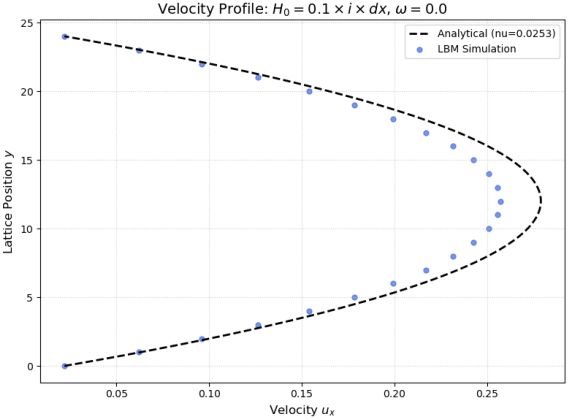
Enhanced velocity profile showing negative viscosity effect.

Magnetic Driven Flow (Zero Pressure Gradient)

- **Kelvin Force Formulation:** The primary driving mechanism is given by the Kelvin body force:

$$\mathbf{F}_K = \mu_0(\mathbf{M} \cdot \nabla)\mathbf{H}$$

- **Mechanism:** Fluid motion is initiated and sustained *purely* by a ramping magnetic field acting on the nanoparticles, with zero external pressure gradient.
- **Implication:** This direct conversion of a ramping magnetic field gradient into bulk translational momentum via the Kelvin force holds high potential for valveless, non-mechanical micro-pumps.



Velocity profile driven entirely by Kelvin magnetic forcing.

Future Work

Antisymmetric Field Application

Investigating complex flow patterns generated solely through specially tailored antisymmetric magnetic fields.

Purely Driven Flow

Developing advanced pumping mechanisms driving flow using only an antisymmetric oscillating field.

Combined Oscillating Fields

Analyzing the coupled effect on apparent viscosity when subjected to both oscillating shear and an oscillatory magnetic field at the same frequency (ω).

Thermal Coupling (Theory)

Formulating the pure theoretical framework for temperature-dependent magnetization and thermal runaway effects in magnetic fluids.

References

1. Shliomis, M. I. (2002). *Effective viscosity of magnetic fluids*. Physical Review E, 64(6).
2. Bacri, J.-C., & Perzynski, R. (1995). *"Negative-Viscosity" Effect in a Magnetic Fluid*. Physical Review Letters, 75(11).
3. Hu, Y., & Li, D. (2018). *Phase-field-based lattice Boltzmann model for multiphase ferrofluid flows*. Physical Review E, 98, 033301.

Thank You